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Multi-Objective Trajectory Optimization for Rigid-Flexible Coupling Spray-Painting Robot Integrated with Coating Process Constraints

Feng Xu¹, Bin Zi^{1,2*} , Jingyuan Wang¹ and Zhaoyi Yu¹

Abstract

Robot-automated spraying is widely used in various fields, such as the automotive, metalworking, furniture, and aerospace industries. Spraying quality is influenced by multiple factors, including robot speed, acceleration, end-effector trajectory, and spraying process constraints. To achieve high-quality spraying under the influence of multiple factors, this study proposes a multi-objective optimization method for the spraying trajectory that integrates spraying process constraints into the optimization process. First, a 7-degree-of-freedom rigid-flexible coupling serial spray painting robot system is introduced, which includes a motion decoupling mechanism and a tension amplification mechanism. Subsequently, a paint deposition model for the spray gun was established, and the influence of process constraints on spraying quality was analyzed. Trajectory planning for the spray painting robot, based on the septic B-spline interpolation method, was then performed. Based on this foundation, objective functions and constraint equations for spraying trajectory optimization were established. A multi-objective trajectory optimization method for spraying by the robot is proposed based on the NSGA-II, which integrates the spraying process constraints. Finally, a prototype system of a 7-degree-of-freedom rigid-flexible coupling serial spray painting robot was constructed. Simulations and spraying experiments were conducted to verify the effectiveness of the proposed multi-objective trajectory optimization method. This paper presents a multi-objective optimization method for the spraying trajectory of a robot. In the proposed method, the optimized spraying trajectory is generated with the spraying process as the constraint and time, energy consumption, and impact during the spraying operation of the robot as the optimization objectives.

Keywords Rigid-flexible coupling spray-painting robot, Painting process, Multi-objective optimization, Trajectory planning, NSGA-II

1 Introduction

Spraying is a crucial process in industrial manufacturing, and high-quality surface spraying endows products with functions such as aesthetic and corrosion protection [1]. Currently, robotic spraying is widely applied in

various fields, including automotive, hardware, furniture, and sanitary ware, with trajectory planning often relying on methods such as manual teaching and offline programming [2]. Manual teaching relies on human experience and requires workers to enter a work environment, leading to issues such as low spraying efficiency, poor stability, and potential health hazards. Existing offline programming methods typically combine the geometric features of a workpiece with slicing algorithms to develop a technical route for spray trajectory planning. However, these methods often fail to consider multiple factors, such as the characteristics of robot motion and the intricacies of the spraying process simultaneously.

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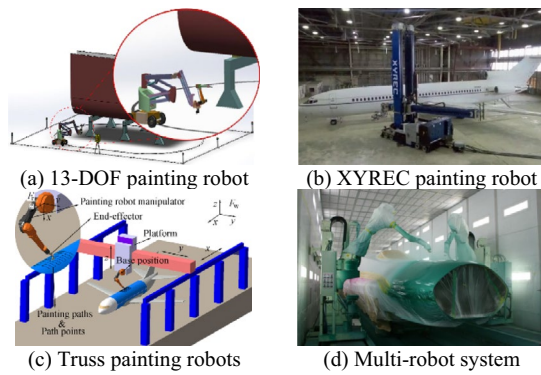


Figure 1 Large-scale rigid painting robot [4–6]

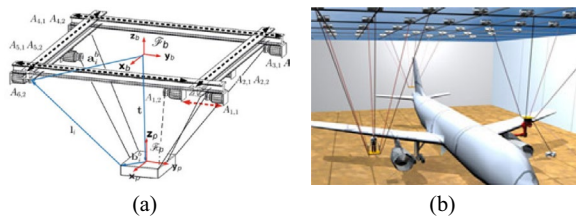


Figure 2 Large-scale cable-driven painting robot [2, 7]: (a) Structural diagram of cable-driven robot, (b) Cable-driven parallel painting robot for aircraft

Furthermore, existing spray robots are predominantly rigid serial ink manipulators [3]; this hinders their applicability for expanding into large-scale equipment. Therefore, the design of a rigid-flexible coupling spray painting robot, along with the proposal of a spraying trajectory optimization method that considers the spraying process, holds significant scientific research significance and engineering application value.

Owing to the immense size and complex structure of large-scale equipment, such as aircraft, ships, high-speed trains, and carriers, automated spraying on their surfaces often involves several methods. These methods include increasing the number of rigid robot joints [4], laying track gantries [5], and employing multirobot collaborative operations [6], as illustrated in Figure 1. However, for large-scale equipment-spraying tasks, traditional rigid robots require advanced track laying, which leads to issues such as high weight, high cost, and low flexibility. Therefore, to achieve a lightweight design for large-scale painting robots and enhance operational flexibility, scholars have developed large-space flexible-driven parallel-spraying robots [7] and used them for the surface coating of large aircraft, as depicted in Figure 2. Although the flexible driving approach enhances the flexibility and workspace of the robots, it also results in a reduction in structural stiffness, motion stability, and end-effector precision. This compromises the ability of large-scale

high-end equipment to satisfy the high-efficiency and high-precision spray requirements.

To meet the aforementioned requirements, scholars have developed a rigid-flexible hybrid parallel painting robot [8], capable of flexible painting operations with multiple postures on a large scale. Additionally, rigid-flexible coupling serial robots feature lightweight structures and long-distance power transmission, enabling their application in large-scale tasks [9]. However, in the process of transmitting driving forces from the base to the end effector, rigid-flexible coupling serial robots encounter issues of joint motion coupling [10]. Currently, there are two main methods for motion decoupling of rigid-flexible coupling serial robots: algorithmic and mechanism decoupling [11, 12]. The algorithmic decoupling method involves a large computational burden, with room for improvement in real-time performance and stability, and it cannot address the problem of flexible cables becoming entangled during joint rolling rotation. Therefore, in a previous study, the authors conducted research on motion decoupling methods and configuration design, designing a sophisticated mechanism for routing driving cables, thus achieving mechanism decoupling of rigid-flexible coupling robots [13].

When robots engage in spray painting operations, they must track specific motion trajectories in both time and space. Trajectory optimization is essential for reducing the energy consumed by the robot, improving operational efficiency, and enhancing the quality of spray painting [14]. Trajectory optimization methods can be broadly classified into two main categories: constrained single-objective optimization and multi-objective optimization. Single-objective optimization methods consider only a single performance index, such as the shortest time, smooth motion, or minimum energy consumption [15], and pursue an optimal single performance index. However, multi-objective optimization methods consider the impact of multiple performance metrics on robot operations simultaneously. By optimizing the balance between these objectives, these methods can achieve optimal trajectories [16]. In recent years, in addition to mathematical optimization algorithms such as the main objective method, coordinated curve method, and unified objective method [17, 18], intelligent optimization algorithms have been extensively researched and applied in the field of robotics. Multi-objective robot trajectory optimization based on intelligent optimization algorithms often employs constraints related to the position, velocity, and acceleration of the robot's end effector or joints [19, 20]. Interpolation methods such as high-order polynomials and B-spline curves are employed to generate motion trajectories in the operational or joint space of a robot [21–24]. Then, based on optimization objectives

such as time-energy consumption minimization, time-jerk minimization, or time-energy consumption-jerk optimization [25–27], intelligent algorithms, including genetic algorithms, particle swarm optimization, and artificial immune algorithms [22, 28–30], are employed to generate optimal trajectories that satisfy task requirements. These trajectory optimization methods focus primarily on the kinematic and dynamic performance of the robot, ignoring the process requirements of the operation task. Therefore, to achieve high-quality spraying operations with the robots, it is necessary to incorporate the spraying process as a constraint for obtaining high-quality spraying trajectories.

To meet the requirements of motion decoupling, modularization, and lightweight design of rigid-flexible coupling robots and to improve the quality of the spray-painting trajectory, this study attempts to construct an improved rigid-flexible coupling painting robot system and proposes a method for optimization of the spray coating trajectory. The main contributions and technical advancements are summarized as follows. Improved cable routing mechanisms and a redundant rigid-flexible coupling series spraying robot system with a motion-decoupling mechanism were developed based on previous research and designs. Furthermore, the spray gun coating deposition model was analyzed, a coating thickness constraint function was established, and a multi-objective trajectory optimization method for spraying robots by integrating the spraying process constraints was proposed. Finally, a prototype system of the redundant rigid-flexible coupling serial spray painting robot was constructed, and painting experiments were conducted.

The remainder of this paper is organized as follows. Section 2 describes the rigid-flexible coupling spray painting robot system. Section 3 presents a spraying trajectory optimization method that considers the process constraints. Optimization simulations and experiments on the spray-painting trajectory are presented in Sections 4 and 5, respectively. The conclusions and future work are presented in Section 6.

2 System Description

To address the motion coupling challenges inherent in the rigid-flexible coupling serial robots, a motion decoupling method is proposed to specifically address the distinct rotational modes of rolling, pitching, and yawing. This approach is based on the translation and rotation of a movable pulley around the axis to achieve effective motion decoupling. Subsequently, a 7-DOF rigid-flexible coupling serial spraying robot system was designed [13], as illustrated in Figure 3.

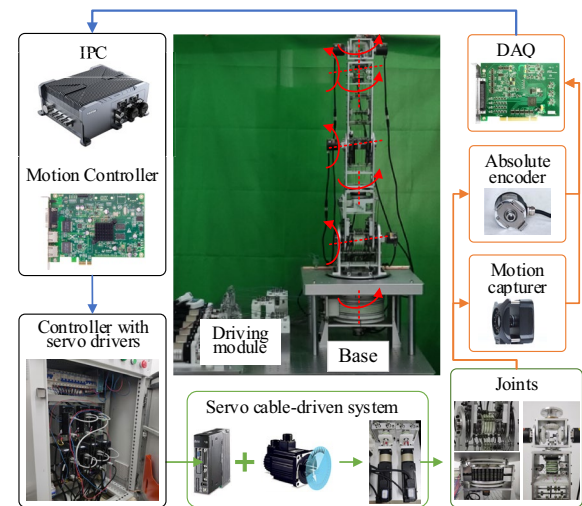


Figure 3 Redundant rigid-flexible coupling spray-painting robot system

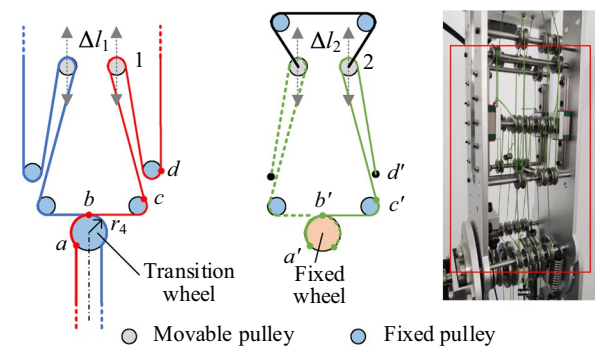


Figure 4 Passive decoupling mechanism based on translational compensation

The robot can be divided into a humanoid shoulder joint (3 DOFs), elbow joint (1 DOF), and wrist joint (3 DOFs). These joints are arranged in pairs along the vertical axis from bottom to top, forming a typical non-biased spherical-roll-spherical configuration. Seven sets of servo actuators are centrally arranged on the base, and the driving forces are transmitted to the seven rotational joints of the robot through several flexible cables. Consequently, the weight of the robot body, excluding the drive mechanism, is reduced, thereby lowering the overall inertia. Simultaneously, two decoupling mechanisms were designed, translational and rotational, to address the issue of motion coupling.

(1) Figure 4 illustrates the design of the mechanism based on the translational compensation passive decoupling method. The core principle of the translational compensation decoupling method involves altering the length of the driving cable passing over a movable pulley by moving it in a fixed direction. This compensates

for the change in the driving cable length over the movable pulley of the joint, ensuring that the total length of the driving cable remains constant and achieves decoupling of the joint motion. In Figure 4, the movable pulley block of joint 4 includes the transition pulley of the driving cable and the fixed pulley of the decoupling cable. Both pulleys are coaxial and have the same diameter. Taking the red driving cable as example, the segment length l_{bc} remains constant because the relative position of the blue-fixed pulley axis remains unchanged. When Link 4 rotates clockwise $\Delta\theta$ around Joint 4, the segment length l_{ab} increases by $\Delta\theta r_4$. To maintain the length (ad segment) of the red driving cable constant during the clockwise rotation of Link 4, ensuring the decoupling of Joint 4 motion, the segment length l_{cd} must decrease by $\Delta\theta r_4$. This necessitates the downward sliding of the gray movable Pulley 1 by Δl_1 . At the same time, the fixed pulley, coaxial and co-diameter with the transition pulley of Joint 4 did not undergo a relative rotation. Consequently, segment $a'b'$ of the green decoupling cable increased by $\Delta\theta r_4$, causing segment $c'd'$ to decrease by $\Delta\theta r_4$, pulling the decoupling movable Pulley 2 downward by Δl_2 . Because the decoupling Pulleys 1 and 2 are coaxial and co-diameter, $\Delta l_1 = \Delta l_2$, that is $\Delta l_{cd} = \Delta l_{c'd'} = -\Delta\theta r_4$. This ensures that the total length of the ad segment of the driving cable remains constant, achieving the decoupling of the rotational motion of Joint 4.

(2) Figure 5 illustrates the design of the mechanism based on the rotational compensation active decoupling method. The core principle of this method involves altering the length of the driving cable passing over a movable pulley by rotating the decoupling pulley around a fixed axis. This compensates for the change in the length of the driving cable over the driving pulley, ensuring that the total length of the driving cable remains constant and achieves the decoupling of the joint motion. Figure 5(a) illustrates the rotational compensation decoupling principle for a single driving cable. Taking the red driving

cable on the right as example, segment A is closer to the base, and the driving and decoupling pulley blocks are coaxial. When the driving pulley is rotated clockwise by $\Delta\theta$, the length l_{DE} of segment DE increases by $\Delta\theta r_1$. To ensure that the total length of segment AF remains constant, the decoupling pulley block must rotate clockwise by $\Delta\theta'$, causing the length of segment BE to decrease by $\Delta\theta r_1$, i.e., $\Delta\theta r_1 = \Delta\theta'(r_1 + r_2)$. Thus, by rotating the decoupling pulley block clockwise using $\Delta\theta'$, the joint motion can be decoupled. Figure 5(b) depicts a 16-cable decoupling mechanism in which the decoupling wheelset consists of several small pulleys.

(3) In traditional rigid serial robots, wrist joints often integrate the drives within the forearm or at the elbow joint, utilizing synchronous belts to drive the motion of the joint. However, this configuration results in a relatively low wrist driving force and larger end-effector inertia. Therefore, this study introduces a cable-driven wrist joint with three degrees of freedom to address these limitations. The design allows for the integration of the drives at the base of the robot, reducing the end-effector inertia, and simultaneously enhancing the driving force at the end-effector over long distances.

As depicted in Figure 6, the pulleys at the lower end of the three-layered shaft sleeve are driven by cables, enabling the independent motion of the shaft sleeve. Subsequently, the bevel gear fixed at the upper end of the shaft sleeve drove the rotation of the large bevel gears on both sides of the joint. Thus, the independent motion of the shaft sleeve and differential motion of the gears enable three degrees of freedom at the end effector.

3 Spraying Trajectory Optimization Considering Process Constraints

Traditional multi-objective trajectory optimization methods for robots often focus on time, energy, and impact. Under the constraints of velocity, acceleration, and jerk, these methods employ mathematical techniques

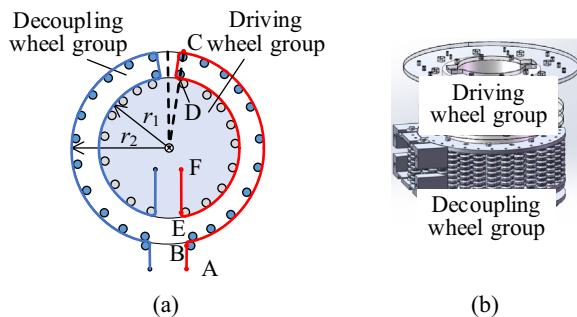


Figure 5 Active decoupling mechanism based on rotational compensation: (a) Principle of rotational compensation, (b) Active decoupling mechanism

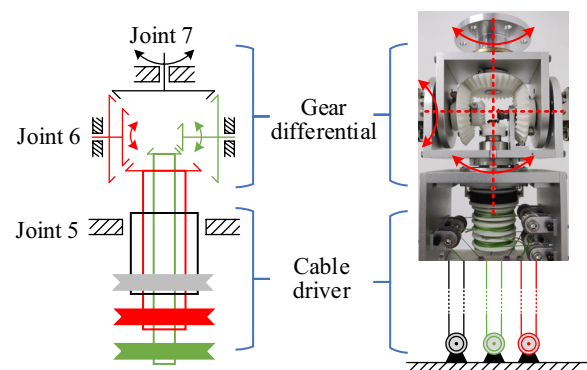


Figure 6 A 3-degree-of-freedom cable-driven wrist joint

or intelligent algorithms to determine the optimal trajectories. However, tasks involving high-quality spraying, robots must not only consider the aforementioned objectives, but also optimize the spraying quality under the constraints of the requirements of the spraying process. The spraying process refers to the surface-finishing process in which paint is atomized by a spray gun and uniformly applied to the workpiece, forming a thin film of paint. The spraying process involves parameters such as the spray distance, spray width, spray gun movement speed, paint viscosity, atomization pressure, air humidity, temperature, and atmospheric pressure [31]. In most cases, robots operate in structured environments in which factors such as air humidity, temperature, and atmospheric pressure remain constant. Before executing the spraying task, parameters such as paint modulation ratio and pressure were pretuned. The aforementioned spraying process and parameters directly affect the quality of the paint film formed after spraying, with the uniformity of the film thickness serving as a crucial criterion for evaluating the spraying quality.

Therefore, this study proposes a multi-objective spraying trajectory optimization method that incorporates the constraints of the spraying process, as depicted in Figure 7. First, paint film thickness served as a process constraint. Second, the spraying trajectories based on septic B-spline curves were generated. Subsequently, a multiobjective function and constraints integrating the spraying process were constructed. Finally, a multi-objective optimization of the spraying trajectories based on NSGA-II was conducted.

3.1 Analysis of Spraying Process and Constraint

Considering the spray distance, spray width, and movement speed of the spray gun as the primary process parameters, the coating deposition model of the spray

gun was analyzed. Subsequently, the thickness of the paint film is designated as a constraint condition for the spraying process, which provides process constraints for the subsequent multi-objective spray trajectory optimization of the rigid-flexible coupling robots.

3.1.1 Coating Deposition Model of Spray Gun

First, a spray gun coating deposition model was established during static spraying, as illustrated in Figure 8 [31]. The rectangular spray gun is an improvement over the circular air spray gun in that the shape of the sprayed paint mist is modified. Therefore, it was necessary to establish a static coating deposition model for the circular spray gun, as indicated by the red spraying area in Figure 8(a).

$$f(r) = C(R^2 - r^2), 0 \leq r \leq R, \quad (1)$$

where r represents the distance between a point on the plane and the center point of the spraying area. C is a constant determined by the parameters of the spraying process such as the spray distance, viscosity, temperature, and pressure. R denotes the radius of the spraying area, which is determined by the spray width. In the Cartesian coordinate system, designating the center point as the origin of the spraying area yields $r = \sqrt{x^2 + y^2}$. Substituting the value of r into Eq. (1) results in the spraying thickness in the Cartesian coordinate system:

$$Z(x, y) = C(R^2 - x^2 - y^2), 0 \leq x^2 + y^2 \leq R^2, \quad (2)$$

The coating ejected from the rectangular spray gun, originally sprayed using a circular spray gun, forms a conical paint mist. After passing through the nozzle plate, a square grey area of side a is formed on the surface of the workpiece. Therefore, by constraining the spraying area, the static coating deposition model of the rectangular spray gun can be obtained using Eq. (3).

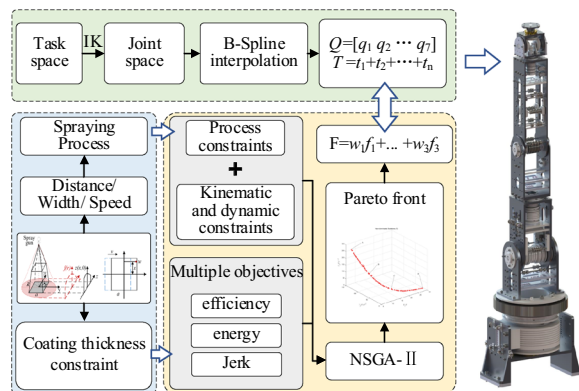


Figure 7 Multi-objective spraying trajectory optimization method integrated with coating process constraints

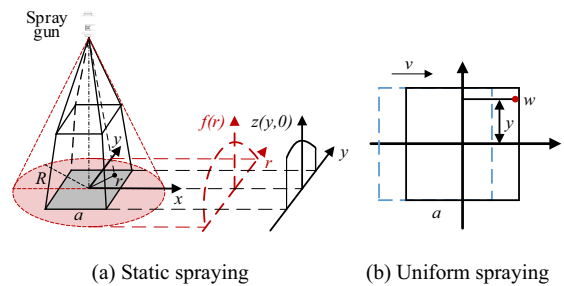


Figure 8 Coating deposition model of rectangular spray gun

$$Z(x, y) = C(R^2 - x^2 - y^2), 0 \leq x \leq \frac{a}{2}, 0 \leq y \leq \frac{a}{2}, \quad (3)$$

where a represents the length of the side of the rectangular spray gun outlet.

Subsequently, a coating deposition model is established for uniform spraying. When a rectangular spray gun is used for camouflage spraying, it is typically oriented perpendicularly to the cutting plane of the workpiece. To ensure stable spraying quality, the spray distance and spray width are generally fixed. The spray gun moves uniformly along the direction of the square edge at velocity v , as shown in Figure 8(b). Let w be a point on the cutting plane in the spraying area. The accumulated coating thickness at this point can be calculated as

$$\delta(y, v) = \frac{Ca(12R^2 - 12y^2 - a^2)}{12v}, (0 \leq x \leq \frac{a}{2}, 0 \leq y \leq \frac{a}{2}), \quad (4)$$

where t_0 is half of the spraying time at point w , $t_0 = a/2v$. y is the distance from point w to the center of the spraying path, $0 \leq y \leq a/2$. x is the distance from point w to the perpendicular axis of the spraying area, $x = a/2 - vt$.

3.1.2 Coating Thickness Constraint

The thickness of the coating directly affects the quality of the film formation, whereby a high-quality film enhances paint adhesion and anchoring effects and increases paint longevity. Consequently, to achieve aesthetically pleasing and durable high-quality coatings, this study incorporates coating thickness as a constraint on the spray trajectory. Typically, during spray operations, the spray gun remains perpendicular to the cutting plane of the workpiece.

For computational convenience, the coordinate system affixed to the end of the spray gun designates the x -axis as the direction of the spray gun movement, whereas the z -axis extends perpendicularly inward from the workpiece cutting plane. Hence, velocity v in Eq. (4) is equivalent to v_x , as illustrated in Figure 9.

To obtain the thickness constraint of the coating in the time domain, the average thickness at any given moment on the trajectory point must first be derived as:

$$\begin{aligned} \bar{\delta}(v_x) &= \frac{2}{a} \int_0^{\frac{a}{2}} \delta(y, v_x) dy \\ &= \frac{Ca^2(6R^2 - a^2)}{12v_x}, \end{aligned} \quad (5)$$

where C is a constant determined by the spraying process parameters such as the spray distance, viscosity, temperature, and pressure. a represents the length of a side of the rectangular spray-gun outlet. R denotes the radius of the spraying area, which is determined by the spray width.

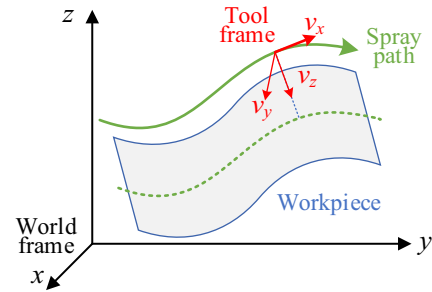


Figure 9 Posture of the spray gun when it moves on the workpiece

The minimum thickness constraint of the coating was determined by Eq. (5).

$$\frac{Ca^2(6R^2 - a^2)}{12v_x} \geq \bar{\delta}_{\min}, \quad (6)$$

where $\bar{\delta}_{\min}$ represents the desired minimum coating thickness.

3.2 Interpolation of Spraying Trajectories Using Septic B-spline

B-spline curves are piecewise polynomial functions with advantages such as geometric invariance, affine invariance, and local support. Therefore, this study adopted the B-spline interpolation method to construct the robot's joint space motion trajectory. The equation for the B-spline function is as:

$$Q(u) = \sum_{j=0}^m B_j^p(u) P_j, u_{\min} \leq u \leq u_{\max}, \quad (7)$$

where $B_j^p(u)$ is the basis function of a p th-degree B-spline curve defined on the knot vector $u = [u_0, \dots, u_{n_{knot}}]$, also known as p th-degree normalized B-spline basis function, which can be determined using the Cox-de Boor recursive formula. Here, $P_j (j = 0, 1, 2, \dots, m)$ represents the multidimensional control points that sequentially connect these control points to form a characteristic polygon.

The continuity between adjacent segments in the B-spline curves is implicitly ensured, making them suitable for addressing one-dimensional problems. Therefore, this study adopts the B-spline interpolation method to construct the joint space motion trajectories of the robot. In this scenario, by setting the node variable $u = t$ in Eq. (7), the B spline function can be expressed as:

$$Q(t) = \sum_{i=0}^m B_j^p(t) P_j, t_{\min} \leq t \leq t_{\max}, \quad (8)$$

where P_j is a scalar parameter that represents the value of the control point at time t .

Given the degree p and the interpolated points θ_k at time $t_k (k = 0, 1, 2, \dots, n)$, the key to trajectory interpolation lies in solving the unknown control points $P_j (j = 0, 1, 2, \dots, m)$ to ensure $Q(t_k) = \theta_k$. Here, we define the node vector $\mathbf{u}$ as:

$$\mathbf{u} = \left[\underbrace{t_0, \dots, t_0}_{p+1}, t_1, \dots, t_{n-1}, \underbrace{t_n, \dots, t_n}_{p+1} \right]. \quad (9)$$

Substituting $n_{knot} - p - 1 = m$, it can be inferred that the number of unknown control points P_j is $m + 1 = (n + 1) + p - 1$. To solve for the unknown coefficients $P_j (j = 0, 1, 2, \dots, m)$, we establish the $n + 1$ equations by ensuring that the interpolation curve passes through θ_k at time t_k :

$$\theta_k = [B_0^p(t_k), B_1^p(t_k), \dots, B_m^p(t_k)] \begin{bmatrix} P_0 \\ P_1 \\ \vdots \\ P_m \end{bmatrix}, k = 0, \dots, n. \quad (10)$$

To obtain a unique solution, $p - 1$ equations are required in addition to the aforementioned $n + 1$ equations. Setting the velocity, acceleration, and jerk (i.e., the i th-order derivatives of the curve) at the starting and ending points as kinematic constraints serves as the boundary condition.

$$\begin{aligned} Q^{(1)}(t_0) &= v_0, \quad Q^{(1)}(t_n) = v_n, \\ Q^{(2)}(t_0) &= \alpha_0, \quad Q^{(2)}(t_n) = \alpha_n, \\ Q^{(3)}(t_0) &= J_0, \quad Q^{(3)}(t_n) = J_n. \end{aligned} \quad (11)$$

The constraint equations can be written as:

$$Q^{(d)}(t_k) = [B_0^{p(d)}(t_k), B_1^{p(d)}(t_k), \dots, B_m^{p(d)}(t_k)] \begin{bmatrix} P_0 \\ P_1 \\ \vdots \\ P_m \end{bmatrix}, \quad (12)$$

where k (0 and n) represents the starting and ending times, respectively. $B_j^{p(d)}(t_k)$ represents the d th-order derivative of the basis function $B_j^{p(d)}(t)$ at time t_k and can be calculated using the following equation:

$$B_j^{p(d)}(t) = \frac{p!}{(p-d)!} \sum_{l=0}^d a_{d,l} B_{j+l}^{p-d}, \quad (13)$$

where

$$\begin{aligned} a_{0,0} &= 1, \\ a_{d,0} &= \frac{a_{d-1,0}}{t_{j+p-d+1} - t_j}, \\ a_{d,l} &= \frac{a_{d-1,l} - a_{d-1,l-1}}{t_{j+p+l-d+1} - t_{j+l}}, \quad l = 1, \dots, d-1, \\ a_{d,d} &= \frac{-a_{d-1,d-1}}{t_{j+p+1} - t_{j+d}}, \end{aligned} \quad (14)$$

For a $p = 7$ th degree B-spline curve, an additional $p - 1$ or p boundary conditions are required to represent the velocity, acceleration, and jerk constraints at the start and end times. In this case, the linear equations can be expressed as:

$$\mathbf{AP} = \mathbf{C}. \quad (15)$$

Expressed in matrix form, it becomes:

$$\begin{bmatrix} B_0^7(t_0) & B_1^7(t_0) & \dots & B_m^7(t_0) \\ B_0^{(1)}(t_0) & B_1^{(1)}(t_0) & \dots & B_m^{(1)}(t_0) \\ B_0^{(2)}(t_0) & B_1^{(2)}(t_0) & \dots & B_m^{(2)}(t_0) \\ B_0^{(3)}(t_0) & B_1^{(3)}(t_0) & \dots & B_m^{(3)}(t_0) \\ B_0^7(t_1) & B_1^7(t_1) & \dots & B_m^7(t_1) \\ \vdots & \vdots & \ddots & \vdots \\ B_0^7(t_{n-1}) & B_1^7(t_{n-1}) & \dots & B_m^7(t_{n-1}) \\ B_0^{(3)}(t_n) & B_1^{(3)}(t_n) & \dots & B_m^{(3)}(t_n) \\ B_0^{(2)}(t_n) & B_1^{(2)}(t_n) & \dots & B_m^{(2)}(t_n) \\ B_0^{(1)}(t_n) & B_1^{(1)}(t_n) & \dots & B_m^{(1)}(t_n) \\ B_0^7(t_n) & B_1^7(t_n) & \dots & B_m^7(t_n) \end{bmatrix} \begin{bmatrix} P_0 \\ P_1 \\ \vdots \\ P_{m-1} \\ P_m \end{bmatrix} = \begin{bmatrix} \theta_0 \\ v_0 \\ \alpha_0 \\ J_0 \\ \theta_1 \\ \vdots \\ \theta_{n-1} \\ J_n \\ \alpha_n \\ v_n \\ \theta_n \end{bmatrix}. \quad (16)$$

According to Eq. (16), the unique control point parameters can be determined by substituting them into Eq. (8), and the expression for the B-spline curve function can be obtained, which represents the interpolated trajectory curve of the septic B-spline in joint space.

After interpolating the joint space trajectory, the first-, second-, and third-order derivatives of the curve representing the velocity, acceleration, and jerk of the joints, during motion, are calculated. The formulas is as follows:

$$Q^{(d)}(t) = \sum_{j=0}^m B_j^{p(d)}(t) q_j, t_{\min} \leq t \leq t_{\max}, \quad (17)$$

where $B_j^{p(d)}(t)$ can be determined using Eq. (13) and q_j represents the derivative control points. The control points for the first derivative are obtained using the following equation:

$$q_j = p \frac{p_{j+1} - p_j}{u_{j+p+1} - u_{j+1}}. \quad (18)$$

The control points for the higher-order derivatives can be obtained recursively using Eq. (18).

To facilitate the calculation of acceleration and jerk integration, the B-spline curve must be rewritten in the form of piecewise polynomials as follows:

$$Q(t) = \begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,7} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,7} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,7} \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \\ t^4 \\ t^5 \\ t^6 \\ t^7 \end{bmatrix}, \quad (19)$$

where $a_{i,j}$ ($i = 0, \dots, n-1, j = 0, \dots, 7$) represents the j th coefficient of the i th segment polynomial, which can be obtained by solving Eqs. (8), (17), and (19) simultaneously.

3.3 Objective Functions Integrated with Constraints of Spraying Process

The robot plans the task-space path based on the actual spraying task, selects a sequence of key path points, and solves the inverse kinematics to generate the key joint-space points $P_{ij} = (q_{ij}, t_i)$, where $i = 0, 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. Here, P_{ij} represents the angle of joint j at path point i , m is the number of robot joints, $n+1$ is the number of path points, and t_i is the time at which the joint reaches path point i . The septic B-spline interpolation method was employed to generate the joint motion trajectories of the robot.

The optimization of multiple objectives necessitates balancing two or more conflicting goals to make optimal decisions. This involves coordinating and compromising various optimization objectives to achieve the overall optimum to the best extent possible. The mathematical model for this is formulated as follows:

$$\begin{cases} \min f(x) = \min [f_1(x), f_2(x), \dots, f_N(x)], \\ \text{s.t.}, g(x) \leq 0, h(x) = 0, \end{cases} \quad (20)$$

where $f_N(x)$ represents the objective function to be optimized, $g(x)$ and $h(x)$ denote the constraint functions, and x denotes the decision variables.

For spraying operations using robots, the following three optimization objective functions were established, considering minimization of time, minimization of energy consumption, and maximization of smoothness:

$$S_1 = T = \sum_{i=1}^n \Delta t_i = \sum_{i=0}^n (t_i - t_{i-1}), \quad (21)$$

$$S_2 = \sum_{j=1}^7 \sqrt{\frac{1}{T} \int_0^T (q_j'')^2 dt}, \quad (22)$$

$$S_3 = \sum_{j=1}^7 \sqrt{\frac{1}{T} \int_0^T (q_j''')^2 dt}, \quad (23)$$

where Δt_i represents the execution time of the i th segment of the path and serves as a decision variable for the objective functions. S_1 denotes the total execution time for the robot to move from the initial point to the terminal point and measures the efficiency of the robot's spraying operation. S_2 shows the average acceleration of all the joints of the robot during motion, indicating the energy consumption of the robot. S_3 indicates the average jerk of all the joints of the robot during motion, measuring the smoothness of the trajectory.

The constraints for robot velocity, acceleration, jerk, and coating thickness are as follows:

$$\begin{cases} \max \{q_j'(t)\} \leq v_{j \max}, \\ \max \{q_j''(t)\} \leq a_{j \max}, \\ \max \{q_j'''(t)\} \leq J_{j \max}, \\ \min \{\delta(t)\} \geq \delta_{\min}, \end{cases} \quad (24)$$

where $\min \{\delta(t)\} \geq \delta_{\min}$ is determined by Eq. (6).

3.4 Multi-Objective Spraying Trajectory Optimization based on NSGA-II

The optimization problem of the spraying trajectory of the robot must simultaneously optimize time, energy consumption, and smoothness, which are mutually conflicting and constrained objectives. For example, a shorter time leads to a greater motion impact. Traditional multi-objective optimization methods often use weighted coefficients to assign weights to each objective, transforming the problem into a single-objective optimization problem. However, this method is subject to significant empirical subjectivity, and the topological structure of the weighted objective function is complex. Compared with traditional algorithms, multi-objective intelligent optimization algorithms offer advantages such as flexibility, effectiveness, adaptability, and stability. Therefore, this study used the NSGA-II algorithm for multi-objective spraying trajectory optimization.

3.4.1 Tent Chaotic Initialization of Population

To ensure population diversity, intelligent algorithms often employ random initialization methods to generate the initial population. However, this can lead to an uneven distribution of the initial population, which is not conducive to the rapid convergence of the algorithm. Therefore, to ensure the diversity and uniform distribution of the initial population simultaneously, this study adopted the tent chaotic mapping method to generate the initial population [32], expressed as follows:

$$x_{n+1} = f(x_n) = \begin{cases} x_n / \alpha, & x_n \in [0, \alpha), \\ (1 - x_n) / (1 - \alpha), & x_n \in [\alpha, 1], \end{cases} \quad (25)$$

$$P_{i,\text{dim}} = lb_{\text{dim}} + (ub_{\text{dim}} - lb_{\text{dim}})x_{i,\text{dim}}, i = 1, 2, \dots, NP, \quad (26)$$

where x_n and x_{n+1} are the variables before and after chaotic mapping, α is a random number between 0 and 1, $P_{i,\text{dim}}$ is the i th initialized population individual, and ub_{dim} and lb_{dim} are the upper and lower bounds of the i th decision variable of the individual.

The steps for population initialization are as follows:

- 1) Randomly generate a dim -dimensional vector $x_0 = (x_{0,1}, x_{0,2}, \dots, x_{0,\text{dim}})$, where each component is a random number between 0 and 1.
- 2) Substitute x_0 into Eq. (25) for NP iterations to obtain the tent chaotic mapping sequence $X = \{x_1, x_2, \dots, x_{NP}\}$.
- 3) Substitute the mapped chaotic sequence X into Eq. (26) to obtain the initialized population containing NP individuals.

3.4.2 Adaptive Crossover and Mutation Probabilities

The crossover and mutation probabilities can significantly impact the performance and efficiency of the algorithm. Traditional algorithms often set constant values. Generally, for the initial evolution of the population, where the individual quality is relatively poor, larger crossover and mutation probabilities are required to explore the global solution space as much as possible. However, as the population evolves, the individual quality improves, necessitating adjustments to the crossover and mutation probabilities to reduce their impact on superior parent individuals, thereby increasing the convergence speed of the algorithm.

The adaptive adjustment of crossover and mutation probabilities can protect optimal individuals while accelerating the elimination of inferior crossover individuals, thereby obtaining the optimal solution as quickly as possible. Therefore, to improve the adaptive efficiency of the algorithm, it is necessary to consider the population iteration stage and the non-dominated sorting level of crossover and mutating individuals and implement an adaptive adjustment of crossover and mutation probabilities [28].

The formulas for adaptive crossover probability and adaptive mutation probability are as follows:

$$P_c = 0.8 + \lambda_{c,i} e^{-\alpha G_i} + \alpha \ln(1 + \beta R_i), \quad (27)$$

$$P_m = 0.03 + \lambda_{m,i} e^{-\alpha G_i} + \alpha \ln(1 + \beta R_i), \quad (28)$$

where $\lambda_{c,i}$ and $\lambda_{m,i}$ are the control coefficients for the crossover and mutation probabilities, respectively, with values of 0.18 and 0.11. α and β are positive parameters, having values of 0.01 and 0.05, respectively. G_i represents the generation of the population iterations. R_i represents the non-dominated sorting level of the crossover or mutating individuals in the current iteration of the population. The value of R_i follows these rules: for the crossover operation, R_i is the smaller non-dominated sorting level among the crossover individuals; for the mutation operation, R_i is the non-dominated sorting level of the mutating individuals.

This method can adaptively adjust the crossover and mutation probabilities based on the population iteration count and non-dominated sorting level, thereby improving the convergence speed of the algorithm.

3.4.3 Tent Chaotic Initialization of Population

In this study, the NSGA-II primarily utilizes tent chaotic mapping to generate an initial population and employs adaptive crossover and mutation operators during genetic operations. The main steps are as follows:

- 1) Initialize the basic parameters of the algorithm.
- 2) Initialize the parent population P_0 based on tent chaotic mapping.
- 3) Calculate the fitness functions of all individuals and perform non-dominated sorting on P_0 .
- 4) Select individuals from P_0 and perform adaptive crossover and mutation operations to generate a new offspring population Q_0 .
- 5) Merge the parent population P_0 and the offspring population Q_0 to generate a new parent population $R_t = P_0 \cup Q_0$.
- 6) Perform non-dominated sorting and ranking on all individuals in R_t and select N offspring individuals according to non-dominated levels. If the total number of individuals in the top l layers of the non-dominated sets is exactly N , proceed to Step 7. If the total number N' of individuals in the top l layers of the non-dominated sets is greater than N , calculate the crowding distance for each individual in each non-dominated layer, select individuals based on non-dominance relations and crowding distance, and form a new parent population of size N .
- 7) Perform select, adaptive crossover, and mutation operations on the N selected individuals to generate a new offspring population Q_t .
- 8) Repeat Steps 5–7 until the maximum number of iterations for the population is reached.
- 9) Finally, generate the Pareto optimal solution set and terminate the algorithm.

The flowchart of the NSGA-II is shown in Figure 10.

4 Trajectory Optimization Simulation

In this study, a simulation analysis of multi-objective spraying trajectory optimization was conducted using the widely adopted transverse raster spraying path used in the industry as an example, as illustrated in Figure 11. The yellow region in the figure denotes the area to be sprayed, where a represents the path spacing.

4.1 Septic B-Spline Trajectory Interpolation

A horizontal straight-line spraying trajectory was selected from Figure 11 as an example, with key path points in the task space chosen as interpolation trajectory points, including the starting, ending, and intermediate points, as shown in Table 1. At each of the eight trajectory points, the robot end-effector was oriented

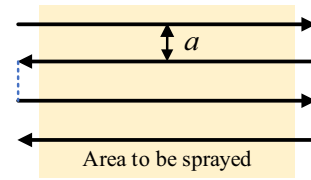


Figure 11 Transverse raster spraying path

vertically to the workpiece to be sprayed, with the end-effector posture represented by fixed XYZ Euler angles as $\alpha = 0$, $\beta = \pi/2$, and $\gamma = 0$, while maintaining the pose unchanged.

Based on the position and orientation information of the aforementioned waypoints, inverse kinematics were solved to obtain the angular values of the seven joints in the joint space, as listed in Table 2.

The times corresponding to the eight trajectory points are defined as $t_0 = 0$ s, $t_1 = 3$ s, $t_2 = 6$ s, $t_3 = 9$ s, $t_4 = 12$ s, $t_5 = 15$ s, $t_6 = 18$ s, and $t_7 = 21$ s. By defining the times of the eight sets of waypoints as the horizontal axis, each waypoint is substituted into Eq. (16) to obtain the inverse solution, resulting in the coordinates of the control points of the septic B-spline curve. Subsequently, by substituting into the parametric equation expressions, the septic B-spline trajectories of each joint passing through the eight interpolation points were obtained. The curves showing the angles, velocities, accelerations, and jerks of the seven joints after septic B-spline interpolation are illustrated in Figures 12, 13, 14 and 15.

4.2 Multi-Objective Spraying Trajectory Optimization

The parameters of the NSGA-II algorithm are set as follows: population size of 200, maximum number of iterations of 50, initial crossover probability of 0.7, and initial mutation probability of 0.05. Using the time intervals Δt_i for each path as decision variables, three optimization objectives—time, energy, and impact—are defined.

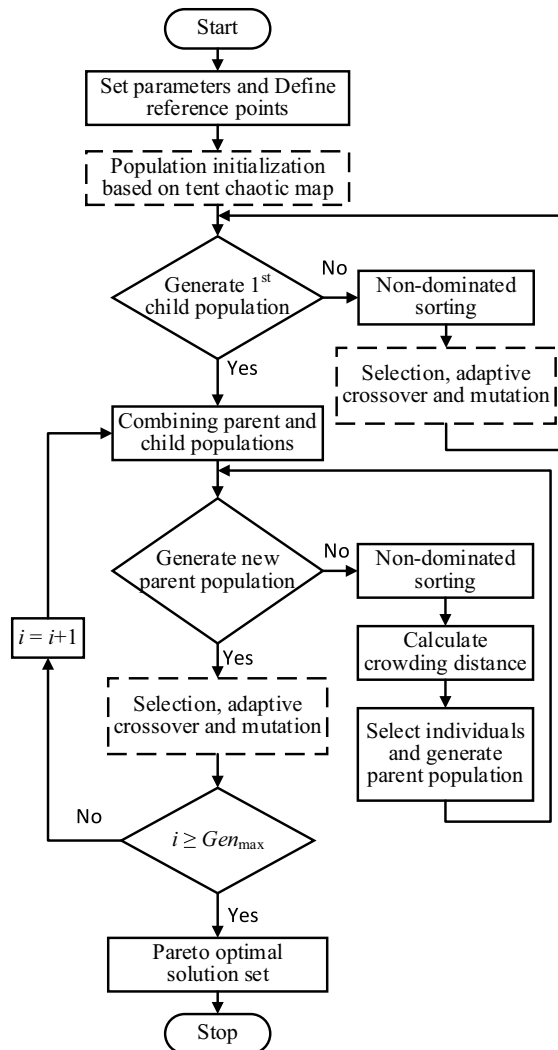


Figure 10 Flowchart for NSGA-II

Table 1 Waypoints in the Cartesian space of the straight-line spraying trajectory

Path points	x (mm)	y (mm)	z (mm)
1	450	− 100	500
2	450	− 70	500
3	450	− 40	500
4	450	− 10	500
5	450	20	500
6	450	50	500
7	450	80	500
8	450	110	500

Table 2 Waypoints in the joint space

Joints	Point 1 (°)	Point 2 (°)	Point 3 (°)	Point 4 (°)
Joint 1	− 14.38	− 10.18	− 5.86	− 1.47
Joint 2	31.45	31.87	32.14	32.25
Joint 3	0	0	0	0
Joint 4	− 9.22	− 10.62	− 11.53	− 11.95
Joint 5	15.48	10.90	6.25	1.57
Joint 6	68.49	69.10	69.51	69.70
Joint 7	− 5.80	− 3.93	− 2.20	− 0.54
Joints	Point 5 (°)	Point 6 (°)	Point 7 (°)	Point 8 (°)
Joint 1	2.94	7.31	11.59	15.75
Joint 2	32.23	32.06	31.75	31.28
Joint 3	0	0	0	0
Joint 4	− 11.86	− 11.28	− 10.21	− 8.64
Joint 5	− 3.13	− 7.81	− 12.44	− 16.99
Joint 6	69.66	69.40	68.92	68.25
Joint 7	1.09	2.76	4.54	6.46

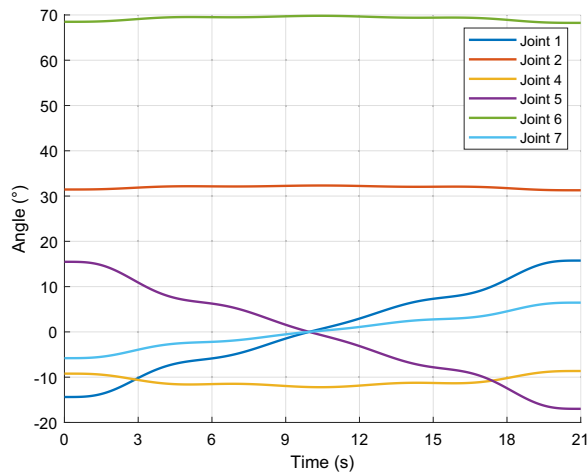


Figure 12 Angle variation for Joint 1–Joint 7

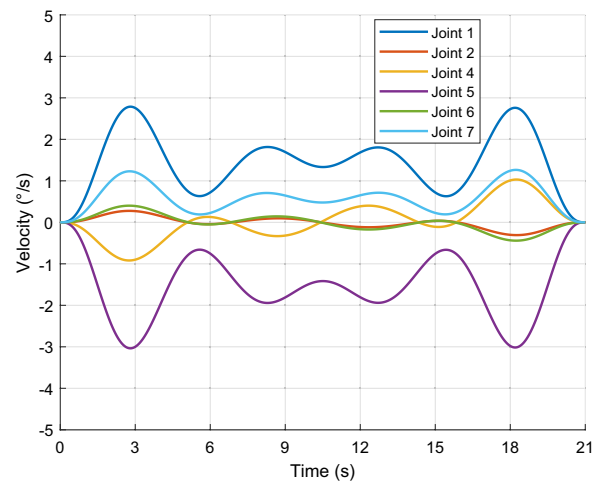


Figure 13 Velocity variation for Joint 1–Joint 7

Under the constraints of velocity, acceleration, jerk, and coating thickness, multi-objective optimization is performed on the spraying trajectory of the rigid-flexible coupling spraying robot. The NSGA-II is executed to optimize the spraying trajectory, resulting in the Pareto optimal solution set shown in Figure 16.

From Figure 16, it can be observed that in the multi-objective trajectory optimization, the optimal performance in terms of impact and energy consumption is inversely proportional to the optimal performance in terms of time. Conversely, the optimal energy consumption is directly proportional to the optimal impact performance. Proximity to point A indicates better time performance in trajectory planning, whereas proximity to

point B indicates better performance in terms of impact and energy consumption in trajectory planning.

The Pareto front is a set of non-dominated solutions. In order to select a unique solution from this set, the weighted objective function is set as $f = \lambda_1 S_1 + \lambda_2 S_2 + \lambda_3 S_3$ to choose a single-objective solution. Using $\lambda_1 = 0.2$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.3$, the optimal solutions are determined as $S_1 = 17.67$, $S_2 = 0.42$, and $S_3 = 0.79$, respectively, resulting in the generation of the optimal spraying trajectory.

The optimized time segment vector is given as $t = [0, 4.02, 6.19, 8.01, 10.14, 12.52, 14.04, 17.67]$. The node vector is $u = [0, 0, 0, 0, 0, 0, 0, 0, 4.02, 6.19, 8.01, 10.14,$

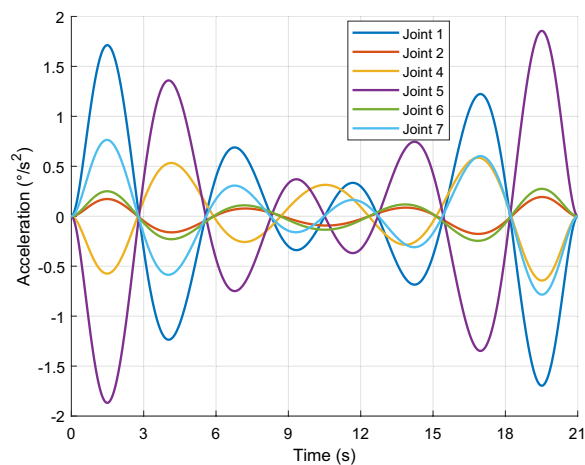


Figure 14 Acceleration variation for Joint 1–Joint 7

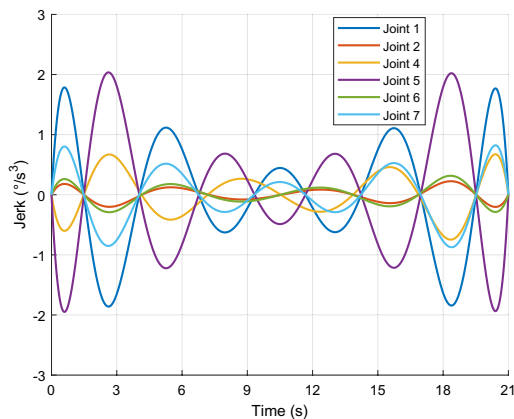


Figure 15 Jerk variation for Joint 1–Joint 7

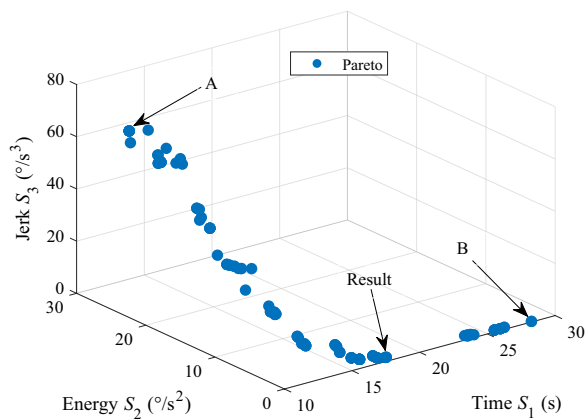


Figure 16 Optimal Pareto front

Table 3 Comparison of time series before and after optimization

Time intervals	Before (s)	After (s)
Δt_1	3	4.02
Δt_2	3	2.17
Δt_3	3	1.82
Δt_4	3	2.13
Δt_5	3	2.38
Δt_6	3	1.52
Δt_7	3	3.63
Total T	21	17.67

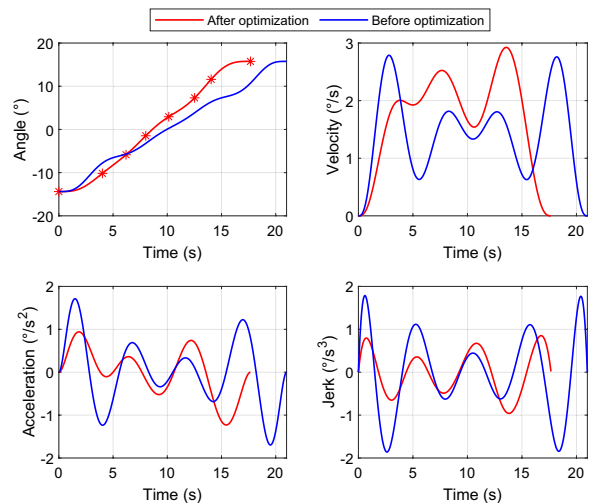


Figure 17 Angle, velocity, acceleration, and jerk of Joint 1 before and after optimization

12.52, 14.04, 17.67, 17.67, 17.67, 17.67, 17.67, 17.67, 17.67, 17.67]. By substituting these into Eq. (19), the angle, velocity, acceleration, and jerk of the robot after the septic B-spline interpolation can be calculated. The time series before and after optimization are listed in Table 3.

The motion curves of each joint after optimization are shown in Figures 17, 18, 19, 20, 21 and 22.

The position, velocity, acceleration, and jerk profiles are depicted in the figures above, and the average velocity, acceleration, and jerk are listed in Table 4.

- 1) A comparison between the optimized and pre-optimized trajectories revealed a reduction in the required time from 21 s to 17.67 s, indicating a partial optimization in time. The execution efficiency increased by 15.9%. Additionally, both the post- and pre-optimization trajectories successfully fulfilled the respective trajectory planning tasks set by the designated planning nodes.

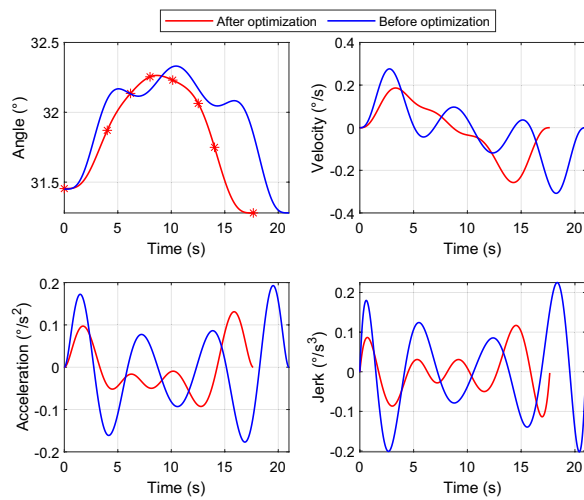


Figure 18 Angle, velocity, acceleration, and jerk of Joint 2 before and after optimization

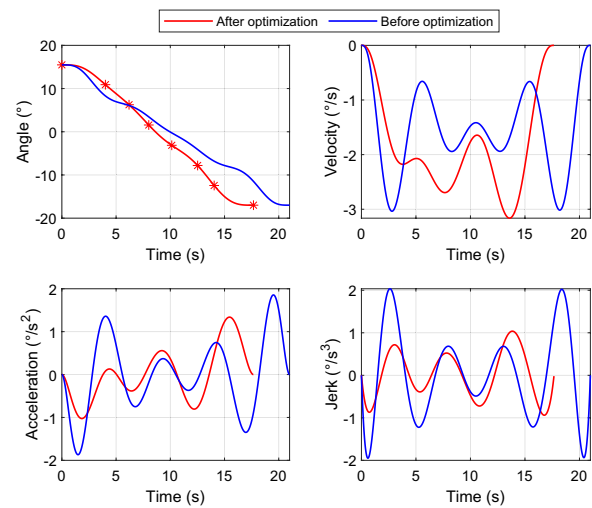


Figure 20 Angle, velocity, acceleration, and jerk of Joint 5 before and after optimization

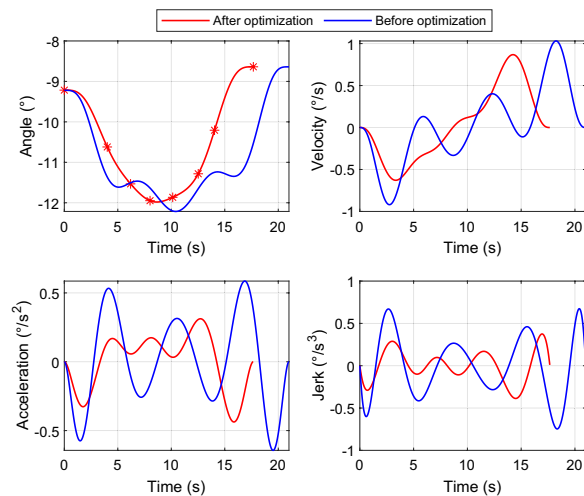


Figure 19 Angle, velocity, acceleration, and jerk of Joint 4 before and after optimization

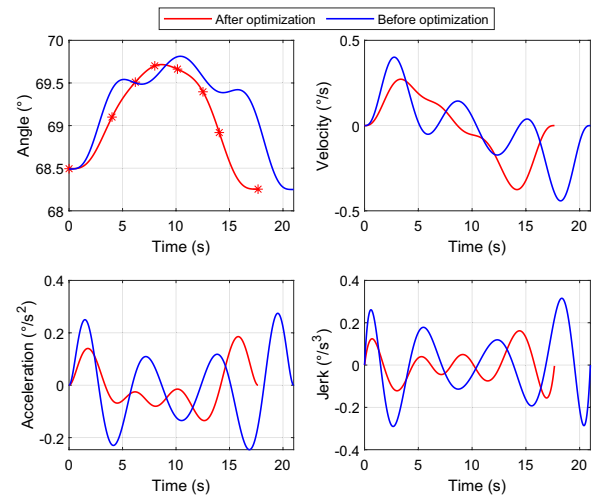


Figure 21 Angle, velocity, acceleration, and jerk of Joint 6 before and after optimization

2) Analysis of the velocity-time and acceleration-time profiles indicates that following optimization, each joint exhibits commencement of motion with a relatively high acceleration. This ensures that the robot achieves higher velocities within shorter durations, thereby enhancing its execution efficiency. The average acceleration decreased from $0.347^\circ/\text{s}^2$ to $0.238^\circ/\text{s}^2$, a reduction of 31.4%.

3) Examination of the jerk-time profile revealed that the average jerk for all joints had effectively improved, resulting in a significant reduction in the

average jerk compared with the pre-optimization. The average jerk decreases from $0.409^\circ/\text{s}^3$ to $0.229^\circ/\text{s}^3$, a reduction of 44%.

In summary, under constraints encompassing the angle, velocity, acceleration, jerk, and thickness, the proposed spraying trajectory optimization method achieved a balanced optimization of efficiency, energy, and impact. Subsequently, practical spraying experiments are conducted to verify whether the optimized spraying trajectories satisfy the actual coating thickness constraints.

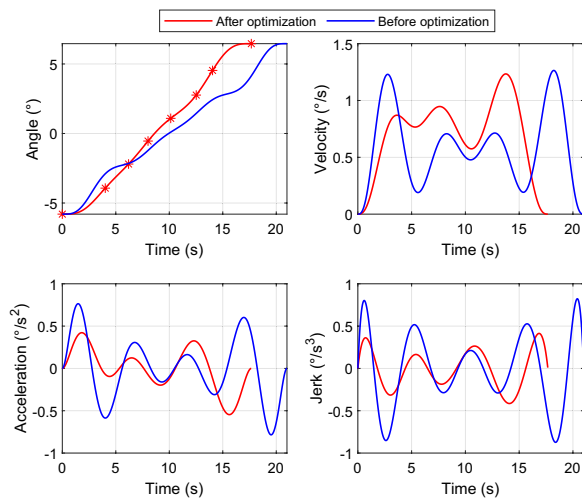


Figure 22 Angle, velocity, acceleration, and jerk of Joint 7 before and after optimization

Table 4 The average velocity, average acceleration, and average jerk of each joint before and after optimization

Joints		Average velocity (°/s)	Average acceleration (°/s ²)	Average jerk (°/s ³)
1	Before	1.435	0.626	0.754
	After	1.705	0.451	0.441
2	Before	0.101	0.084	0.091
	After	0.102	0.05	0.045
4	Before	0.337	0.279	0.304
	After	0.346	0.169	0.15
5	Before	1.546	0.686	0.825
	After	1.838	0.489	0.48
6	Before	0.146	0.119	0.13
	After	0.152	0.073	0.064
7	Before	0.584	0.291	0.35
	After	0.694	0.194	0.194
All joints	Before	0.691	0.347	0.409
	After	0.806	0.238	0.229

5 Spraying Experiment

The primary objective of the spraying experiment was to verify whether the spraying trajectory planned using the aforementioned method satisfied the paint thickness constraint. Accordingly, an experimental platform for the redundant rigid-flexible coupling spraying robot system was constructed, which included a rigid-flexible coupling spraying robot system and an automatic paint spraying system, as shown in Figure 23.

The automatic spraying system consisted of an air compressor, diaphragm pump, filter pressure-reducing

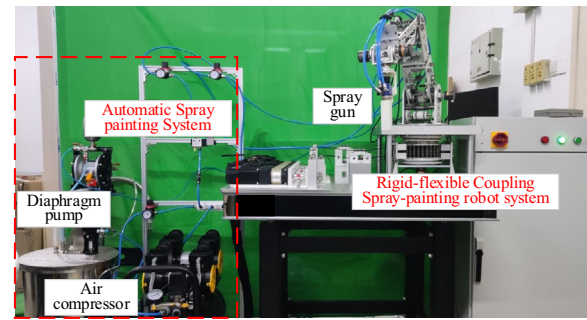


Figure 23 Experimental platform of robot spraying system

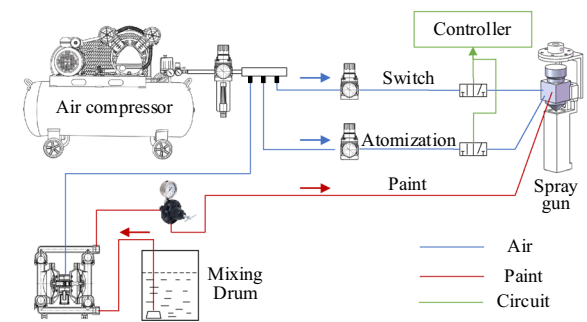


Figure 24 Automatic spray-painting system

valve, stabilizing valve, solenoid valves, and a rectangular spray gun, as shown in Figure 24.

Before conducting the spraying experiment, the paint spraying process parameters are set as follows: control pressure is 0.2 MPa, atomization pressure is 0.15 MPa, and feeding pressure of 0.1 MPa. The flow rate is controlled by the top knob of the spray gun to ensure sufficient atomization and the absence of droplets. Planar and curved surface-spraying experiments were performed to verify the effectiveness of the trajectory optimization method.

(1) A spraying experiment is conducted using a planar workpiece. After optimizing the spraying trajectory using the NSGA-II, a spraying experiment was conducted on the planar workpiece, as depicted in Figure 25.

The spraying process for executing the first trajectory is illustrated in Figure 25, where (a) represents the initial moment and (b) depicts the moment at $t = 2$ s. Upon completion of the first trajectory, the spray gun is turned off, after which the robot is moved to the second trajectory, and the spray gun is turned on again to continue spraying. This process was repeated until all spraying trajectories were executed. After completion of the spraying process, the intersection between the jet axis and spraying plane was calibrated. The results of the spraying

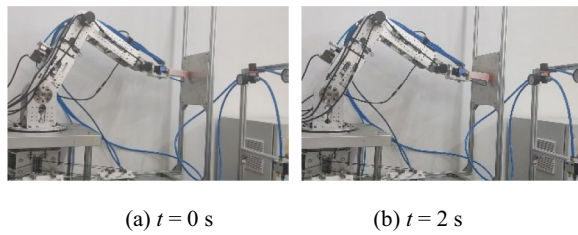


Figure 25 Set-up for spray-painting experiment

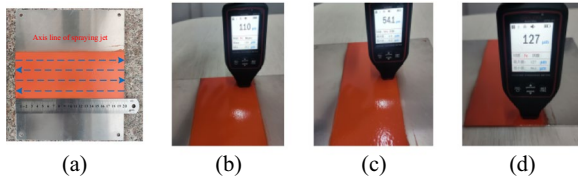


Figure 26 Results of the spraying experiment after optimization: (a) Overall spraying effect, (b) Thickness of paint film near the starting point, (c) Thickness of paint film near the midpoint, (d) Thickness of paint film near the termination point

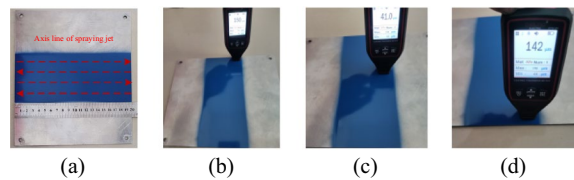


Figure 27 Results of the spraying experiment before optimization: (a) Overall spraying effect, (b) Thickness of paint film near the starting point, (c) Thickness of paint film near the midpoint, (d) Thickness of paint film near the termination point

experiment, after and before optimization, are shown in Figures 26, and 27, respectively.

Figure 26(a) shows a planar workpiece after the optimized spraying. The paint thickness was measured after calibrating the intersection point of the jet axis line with the plane to be sprayed. Figure 26(b), (c), and (d) depict the paint thickness near the initial, middle, and final positions in the first trajectory, respectively. Similarly, Figure 27(a) shows the planar workpiece before optimized spraying. The paint thickness was measured after calibrating the intersection point of the jet axis line with the plane to be sprayed. Figure 27(b), (c), and (d) depict the paint thickness near the initial, middle, and final positions in the first trajectory. Taking the first spraying trajectory as example, the paint film thicknesses at the eight interpolation points before and after optimization were measured, as listed in Table 5.

Analysis of Table 5 reveals that the paint film thickness after optimization on the workpiece surface ranges

Table 5 Actual thickness of paint film on the first spraying trajectory of the planar workpiece before and after optimization

Points	y (mm)	Thickness before and after optimization	
		Before (μm)	After (μm)
1	− 100	144	123
2	− 70	112	93.6
3	− 40	67.9	88.6
4	− 10	41.0	50.3
5	20	47.4	68.2
6	50	75.1	59.1
7	80	109	93.5
8	110	139	133

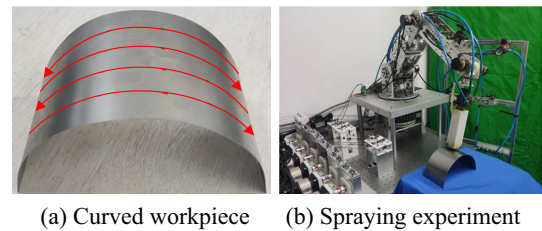


Figure 28 Spraying experiment on semi cylindrical workpiece

between 50.3 μm and 133 μm . The average thickness of the paint film is 88.7 μm . The paint film thickness on the workpiece surface, before optimization, ranges between 41.0 μm and 144 μm . The average paint film thickness was 91.9 μm . An analysis of the above data shows that the thickness measured near the start and end positions exhibit larger values than those near the middle position. This discrepancy can be attributed to the acceleration and deceleration actions of the robot near the start and end moments, whereas the robot's movement is relatively smooth near the middle moment, aligning with the motion pattern of the robot.

(2) A spraying experiment was conducted on a curved workpiece to verify the effectiveness of the optimization method for optimizing the spraying trajectory on curved surfaces. A semi-cylinder was selected as the workpiece for spraying. The red line is the spraying trajectory, as shown in Figure 28(a).

The arc length of the semi cylinder is 200 mm and the diameter is $400/\pi$ mm. One of the arc-spraying trajectories shown in Figure 28(a) is chosen as example, with key path points in the task space chosen as interpolation trajectory points. These include the starting, ending, and intermediate points as shown in Table 6. At each of the eight trajectory points, the robot end effector is oriented vertically to the workpiece to be sprayed

Table 6 Waypoints in the Cartesian space of arc spraying trajectory

Waypoints	Cartesian coordinates (x, y, z) mm	XYZ Euler angles (γ, β, α) rad
1	(380, -60, 50)	($\pi/2, \pi, 0$)
2	(380, -66.3, 77.6)	($5\pi/14, \pi, 0$)
3	(380, -83.9, 99.8)	($3\pi/14, \pi, 0$)
4	(380, -109.5, 112.1)	($\pi/14, \pi, 0$)
5	(380, -137.8, 112.1)	($-\pi/14, \pi, 0$)
6	(380, -163.4, 99.8)	($-3\pi/14, \pi, 0$)
7	(380, -181.0, 77.6)	($-5\pi/14, \pi, 0$)
8	(380, -187.3, 50)	($-\pi/2, \pi, 0$)

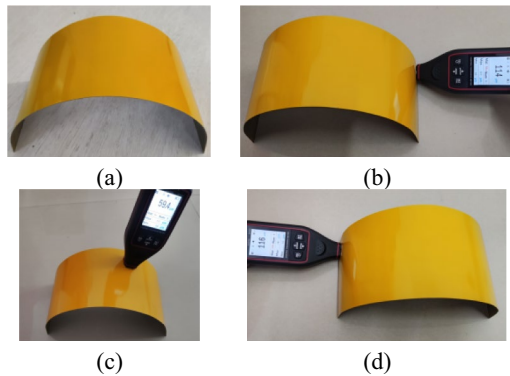


Figure 29 Results of the spraying experiment on the curved workpiece after optimization: (a) Overall spraying effect, (b) Thickness of paint film near the starting point, (c) Thickness of paint film near the midpoint, (d) Thickness of paint film near the termination point

with the end-effector posture represented by XYZ Euler angles which are fixed.

The inverse kinematics of the robot were solved to determine the joint interpolation points corresponding to the interpolation points in the Cartesian space listed in Table 6: Subsequently, a seventh-order B-spline interpolation method is employed to obtain the motion trajectory in the joint space. The proposed multi-objective optimization method is then utilized to generate a spray trajectory based on the constraints of the spraying process. Specific calculation examples have been provided earlier and are not reiterated here. Finally, the robot executes the spraying trajectories before and after the optimization to conduct the spraying experiments, as illustrated in Figure 28(b). The results of the spraying experiment on the curved workpiece after optimization are shown in Figure 29, and the results of the spraying experiment on the curved workpiece before optimization are shown in Figure 30.

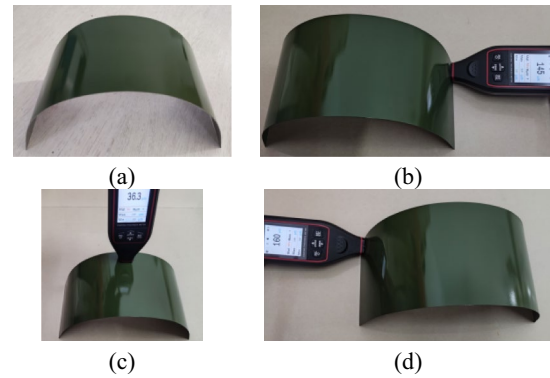


Figure 30 The results of the spraying experiment on the curved workpiece before optimization: (a) Overall spraying effect, (b) Thickness of paint film near the starting point, (c) Thickness of paint film near the midpoint, (d) Thickness of paint film near the termination point

Table 7 Actual paint film thickness on the first spraying trajectory of the curved workpiece before and after optimization

Waypoints	Thickness before and after optimization	
	Before (μm)	After (μm)
1	160	114
2	106	93.6
3	89.9	70.2
4	36.3	59.4
5	52.8	69.1
6	97.2	59.1
7	111	93.5
8	145	126

Figure 29(a) shows the curved workpiece after optimized spraying. The paint thickness was measured after calibrating the intersection point of the jet axis line with the plane to be sprayed. Figure 29(b), (c), and (d) respectively depict the paint thickness near the initial, middle, and final positions in the first trajectory. Similarly, Figure 30(a) shows the curved workpiece before the optimized spraying process. Figure 30(b), (c), and (d) respectively depict the paint thickness near the initial, middle, and final positions in the first trajectory. Taking the first spraying trajectory as example, the paint film thicknesses at the eight interpolation points before and after optimization were measured, and are listed in Table 7.

Analysis of Table 7 reveals that the paint film thickness on the curved workpiece, after optimization, ranges between 59.1 μm and 126 μm . The average thickness of the paint film is 85.6 μm . The thickness of the paint film

on the curved workpiece, before optimization, ranges between 36.3 μm and 160 μm . The average thickness of the paint film is 99.8 μm . Similar to the case of planar spraying, the measured thicknesses at the start and end points were greater than the measured thicknesses near the middle point. This phenomenon is primarily attributed to the zero velocity of the robot at the start and end points, which results in significant acceleration and deceleration near these regions.

A comprehensive analysis of the results of planar spraying experiments and curved surface spraying experiments showed that (1) The thickness of the paint film before optimization was lower than the minimum thickness constraint, and the average thickness was greater. (2) The minimum measured thickness after optimization exceeds 45 μm , and the average thickness on the planar and curved workpieces decreased by 3.6% and 14.2% respectively. This shows that the optimized film thickness satisfies the paint thickness constraint and validates the effectiveness of the proposed spray trajectory optimization method, which integrates the process constraints.

6 Conclusions

In this study, a multi-objective trajectory optimization method for spraying robots that integrates spraying process constraints is proposed, and a redundant rigid-flexible coupling serial spraying robot system is constructed. Simulations and spraying experiments are conducted to verify the effectiveness of the proposed method. The following conclusions were drawn.

- (1) The simulation results before and after optimization show that the required time reduces from 21 s to 17.67 s, and the execution efficiency increases by 15.9%. In addition, the trajectories before and after optimization successfully fulfilled the respective trajectory-planning tasks set by the designated planning nodes.
- (2) After optimization, the robot exhibited an increased acceleration near the starting and ending positions, which reduced the paint thickness. Additionally, the average acceleration decreased from $0.347^\circ/\text{s}^2$ to $0.238^\circ/\text{s}^2$, a reduction of 31.4%. The results indicate that the optimized trajectory reduces energy consumption while enhancing operational efficiency.
- (3) The jerk-time profile reveals that the average jerk is significantly reduced compared to the pre-optimization state. The average jerk decreased from $0.409^\circ/\text{s}^3$ to $0.229^\circ/\text{s}^3$, a reduction of 44%. This indicates that the proposed optimization method effectively reduces the impact on the robot during its motion.

- (4) The results of the spraying experiments after optimization reveal that the paint film thickness on planar and curved workpieces ranges between 50.3 μm to 133 μm and 59.1 μm to 126 μm , respectively. Combined with the findings of the simulation analysis, this indicates that the optimized spray trajectory achieves a smoother motion while satisfying the minimum paint thickness constraint.

The results of the simulations and spraying experiments validated the effectiveness of the proposed method, ensuring that the paint thickness was not less than the minimum expected value while balancing multiple optimization objectives. Future work will consider interpolation methods for complex freeform surfaces and explore coating thickness uniformity as an optimization objective in multi-objective spraying trajectory optimization.

Authors' Contributions

BZ was responsible for the entire trial, FX wrote the manuscript, JW assisted with the experimental setup, and ZY assisted with the structure and language of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing Interests

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